

Blocks with lectures

- I.) Prüfungsfach „Foundations of Medical Informatics“ (30 ECTS)
 - 1.) Modul „Advanced Software Engineering in the Biomedical Context“ (6,0 ECTS)
 - 2.) Modul „Biological Basis of Disease“ (6,0 ECTS)
 - 3.) Modul „Introduction to Medical Data Science and Analytics“ (6,0 ECTS)
 - 4.) Modul „Introduction to Digital Health“ (6,0 ECTS)
 - 5.) Modul „Practice of Medicine“ (6,0 ECTS)
- II.) Prüfungsfach „Biological and Medical Principles“
 - 1.) *Modul „Bioinformatics“ (6,0 ECTS)
 - 2.) *Modul „Computational Neuroscience“ (6,0 ECTS)
 - 3.) *Modul „Introduction to Molecular Biology“ (6,0 ECTS)
 - 4.) *Modul „Introduction to Neuroscience“ (6,0 ECTS)
 - 5.) *Modul „Neuroinformatics“ (6,0 ECTS)
 - 6.) *Modul „Prosthetics and Control Theory“ (6,0 ECTS)
- III.) Prüfungsfach „Data, Policy, and Health Informatics“
 - 1.) *Modul „Advanced Database Systems“ (6,0 ECTS)
 - 2.) *Modul „AI Ethics“ (6,0 ECTS)
 - 3.) *Modul „Data Stewardship“ (6,0 ECTS)
 - 4.) *Modul „Methodological Foundations of Public Health Informatics“ (6,0 ECTS)
 - 5.) *Modul „Public Health Policy and Healthcare Management“ (6,0 ECTS)
 - 6.) *Modul „Responsible Digital Ethics“ (6,0 ECTS)
 - 7.) *Modul „Standardized Health Information Management“ (6,0 ECTS)
- IV.) Prüfungsfach „Machine Learning and Intelligent Health Systems“
 - 1.) *Modul „Advanced Machine Learning in Healthcare“ (6,0 ECTS)
 - 2.) *Modul „Medical Signal Processing and Imaging“ (6,0 ECTS)
 - 3.) *Modul „Computer Vision“ (6,0 ECTS)
 - 4.) *Modul „Generative AI, General AI“ (6,0 ECTS)
 - 5.) *Modul „Logic-based Artificial Intelligence“ (6,0 ECTS)
 - 6.) *Modul „ML in Medical Imaging and Precision Medicine“ (6,0 ECTS)
 - 7.) *Modul „Modeling and Simulation for Medical Informatics“ (6,0 ECTS)
- V.) Prüfungsfach „Human-Centered Technologies and Visualization in Healthcare“
 - 1.) *Modul „Artifact-based Design“ (6,0 ECTS)
 - 2.) *Modul „Assistive Technologies“ (6,0 ECTS)
 - 3.) *Modul „Human-Centered AI“ (6,0 ECTS)
 - 4.) *Modul „Information Visualization“ (6,0 ECTS)
 - 5.) *Modul „Medical Visualization and Visual Analytics“ (6,0 ECTS)
 - 6.) *Modul „Virtual and Augmented Reality“ (6,0 ECTS)
 - 7.) *Modul „Wellbeing Technologies“ (6,0 ECTS)
- VI.) Prüfungsfach „Free Electives“
 - 1.) Modul „Freie Wahlfächer und Transferable Skills“ (9,0 ECTS)
- VII.) Prüfungsfach „Seminar, Project, and Advanced Topics in Medical Informatics“
 - 1.) Modul „Seminar in Medical Informatics“ (3,0 ECTS)
 - 2.) Modul „Project in Medical Informatics“ (6,0 ECTS)
 - 3.) *Modul „Advanced Topics In Medical Informatics“ (min. 3,0 ECTS)
 - 4.) *Prüfungsfach „Masterarbeit“